SMART AUTOMATED WATER TANK LID SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

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MEB1233- UG PROJECT WORK

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Under the Guidance of

Mr. B. VELUCHAMY

Department of Mechanical Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

MECHANICAL ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled "**SMART AUTOMATED WATER TANK LID SYSTEM**""**is** the bonafide work of **GIDEON CYRUS E, (8115U24ME019), HARIHARAN M, (8115U24ME022), KEERTHIVARMAN G, (8115U24ME030), LARSHAD E, (8115U24ME032),** who carried out the work under my supervision.

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DECLARATION BY THE CANDITATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

1.
2.
3.
4.

SIGNATURE OF THE CANDITATE

ACKNOWLEDGMENT

We thank the almighty god without his blessing it would not have been possible for us to complete this project.

At this moment of having successfully completed our project, we wish to convey our sincere thanks and gratitude to our management of our college and our beloved chairman **Dr. K. RAMAKRISHNAN, B.E., Ph.D., who** provide all the facilities to us.

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K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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VISION OF THE K. RAMAKRISHNAN COLLEGE OF ENGINEERING

- To achieve a prominent position among the top technical institutions

MISSION OF THE K. RAMAKRISHNAN COLLEGE OF ENGINEERING

- To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.
- To nurture research and entrepreneurial skills among students in cutting edge technologies
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PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

- **PEO1-** Solve real world problems using advanced computing technologies.
- **PEO 2-** Encompass the ability to build innovative software products and become a successful entrepreneur.
- **PEO 3-** Carry out industrial projects and succeed in research careers.

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PROGRAM OUTCOMES [PO's]

Engineering students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems; design systems/components/processes meeting specified needs with safety, societal, and environmental considerations.
4. **Investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Engineering Tool Usage (Modern tool usage):** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and the world:** A merged outcome combining societal, environmental, sustainability concerns; graduates apply reasoning to assess societal, health, safety, legal, cultural, and environmental impacts of engineering solutions.
7. **Ethics and Inclusive Professional Architecture:** Apply ethical principles and commit to professional ethics, responsibilities, inclusive and value-based behavior in engineering practice.
8. **Individual and Collaborative team work:** Function effectively as an individual, and as a member or leader in diverse and multidisciplinary teams, with inclusive and collaborative behavior.

9. **Communication and Global Awareness:** Communicate effectively on complex engineering activities with the engineering community and society at large: writing reports/design docs, effective presentations, clear instructions; with sensitivity to global and inclusive context.

10. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments consider economic and Financial aspects

11. **Life-long learning Adaptability and Tech Awareness:** Recognize need for, and have preparation and ability to engage in independent and lifelong learning, keeping up with evolving technology, global standards, and professional growth.

PROGRAM SPECIFIC OUTPUTS (PSO'S):

PSO1: To apply the knowledge of computer programming to develop the software projects.

PSO2: To design software in futuristic approach to adapt cutting edge technologies.

PSO3: To solve societal problems using computation intelligence.

PO'S & PSO'S ATTAINMENT: *(Put tick Mark for the attained PO's& PSO's)*

PO1	**Engineering Knowledge**		**PO7**	**Ethics and Inclusive Professional Architecture**		**PSO1**	**Apply Computer Programming**	
PO2	**Problem Analysis**		**PO8**	**Individual and Collaborative team work**		**PSO2**	**Design Software**	
PO3	**Design & Development**		**PO9**	**Communication Skills**		**PSO3**	**Solve societal Problems**	
PO4	**Investigations**		PO10	**Project Mgt. & Finance**				
PO5	**Modern Tools**		**PO11**	**Life Long Learning**				
PO6	**Engineer & Society**							

SUSTAINABLE DEVELOPMENT GOALS ATTAINED: (SDG's)



SDG's ATTAINED:

SDG's *(Put tick Mark for the attained SDG's)*																
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
		✓						✓		✓	✓					

ABSTRACT

The Smart Automated Water Tank Lid System is an innovative solution designed to improve water hygiene, reduce manual effort, and enhance safety in water storage systems. Traditional water tanks often remain open or improperly covered, leading to contamination, evaporation, and the entry of dust, insects, and debris. This project introduces an automated lid mechanism that operates based on water level detection using sensors.

The system integrates a microcontroller, water level sensors, and a motorized lid mechanism to automatically open and close the tank lid when required. When the water level drops below a certain limit, the lid opens to allow refilling, and once the tank reaches its maximum level, the lid closes securely. This automation minimizes human intervention and ensures proper sealing of the tank at all times.

Additionally, the system can be enhanced with IoT features for remote monitoring and alerts, making it suitable for smart homes and industrial applications. The proposed design is cost-effective, energy-efficient, and easy to install, making it a practical solution for modern water management.

Key Words: Hygiene, effortless, allows refilling automatically, automatic refiiling

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CHAPTER 1
INTRODUCTION

1.1 Overview of Design Thinking:

Design thinking is a user-centered approach that focuses on understanding real-world problems and developing practical, innovative solutions. In the Smart Automated Water Tank Lid System, the process begins by identifying common issues faced by users such as open tank lids, water contamination, mosquito breeding, and the need for manual operation. By empathizing with users like homeowners and maintenance personnel, the problem is clearly defined as the need for an automated system that ensures proper tank closure while allowing efficient water filling.

The ideation phase involves generating possible solutions such as using water level sensors, motorized lids, and microcontrollers like Arduino to automate the process. A prototype is then developed by integrating these components into a working model that can automatically open and close the tank lid based on water levels. This model is tested for efficiency, reliability, and ease of use, and improvements are made based on feedback.

Overall, applying design thinking to this system ensures that the final solution is not only technically effective but also user-friendly, cost-efficient, and scalable. It helps create a smart solution that enhances water hygiene, reduces manual effort, and contributes to better water management practices.

1.2 Importance of Design Thinking in Mechanical Engineering:

Design thinking plays a crucial role in mechanical engineering by ensuring that solutions are not only technically sound but also practical and user-focused. In the

Smart Automated Water Tank Lid System, it helps engineers understand real-life challenges such as contamination, manual handling, and inefficient tank management. By focusing on user needs, engineers can design a system that is easy to operate, safe, and reliable in everyday conditions rather than just theoretically effective.

In this project, design thinking guides the selection of appropriate mechanical components such as linkages, hinges, and motor mechanisms, along with control systems like Arduino. It encourages innovation in creating an automated lid that opens and closes smoothly based on water levels while ensuring durability and minimal maintenance. This approach also helps in optimizing material usage, reducing cost, and improving energy efficiency, which are key aspects of mechanical engineering design.

Overall, the importance of design thinking lies in bridging the gap between engineering concepts and real-world application. It ensures that the Smart Automated Water Tank Lid System is efficient, economical, and user-friendly, making it a practical solution for modern water management and smart infrastructure development.

1.3 Project Objectives:

The primary objective of the Smart Automated Water Tank Lid System is to develop an automated mechanism that ensures the water tank remains securely closed when not in use, thereby preventing contamination from dust, insects, and other external elements. The system aims to maintain water hygiene and improve overall safety by eliminating the need for manual lid operation.

Another key objective is to integrate water level sensing with a control unit, such as Arduino, to enable automatic opening and closing of the lid based on the tank's water level. This helps in reducing human effort, preventing overflow or unnecessary exposure, and ensuring efficient water management.

Additionally, the system is designed to be cost-effective, energy-efficient, and easy to

install and maintain. It also aims to provide scalability for future enhancements like IoT-based monitoring and alerts, making it suitable for both domestic and industrial applications.

1.4 Scope and Limitations of the Project:

The Smart Automated Water Tank Lid System is designed to improve water storage hygiene and reduce manual intervention through automation. The project focuses on developing a system that automatically opens and closes the tank lid based on water level detection using sensors and a control unit such as Arduino. It can be applied in residential buildings, apartments, and small-scale industrial setups. The system is scalable, allowing future integration with IoT for remote monitoring, alerts, and smart home connectivity. It also emphasizes energy efficiency, low cost, and ease of installation, making it suitable for practical real-world use.

Despite its advantages, the system has certain limitations. Its performance depends on the accuracy and durability of sensors, which may be affected by water quality, dirt, or environmental conditions. The mechanical components like motors and hinges may require regular maintenance over time. The system also relies on a continuous power supply, which may limit its use in areas with frequent power outages unless supported by backup systems. Additionally, advanced features like IoT integration may increase the overall cost and complexity of the system.

The Smart Automated Water Tank Lid System is highly feasible from multiple perspectives, making it suitable for practical implementation. **Technically**, the system uses readily available components such as water level sensors, motors, and microcontrollers like Arduino, which are easy to integrate and widely supported. The design is simple and does not require complex manufacturing processes, allowing it to be developed and maintained with basic mechanical and electronic knowledge.

From an **economic feasibility** standpoint, the project is cost-effective as it uses low-cost materials and components. The maintenance cost is minimal,

and the long-term benefits such as improved water hygiene and reduced manual labor justify the investment. This makes it affordable for households, apartments, and small industries.

CHAPTER 2

EMPATHIZE – PROBLEM IDENTIFICATION & USER STUDY

2.1 Background of the Problem:

Water storage tanks are widely used in residential, commercial, and industrial settings to ensure a continuous water supply. However, many of these tanks rely on manually operated lids, which are often left open or not properly secured. This leads to several issues such as contamination from dust, insects, and debris, as well as the risk of mosquito breeding, which can affect water quality and public health. In addition, frequent manual checking of water levels and lid operation increases human effort and the chances of negligence.

With increasing awareness of hygiene and the need for efficient resource management, traditional water tank systems are no longer sufficient. There is a growing demand for smart and automated solutions that can maintain cleanliness, reduce manual intervention, and improve overall system efficiency. Technologies such as sensors and microcontrollers like Arduino have made it possible to design systems that respond automatically to changing conditions.

Therefore, the need arises for a Smart Automated Water Tank Lid System that can address these challenges by ensuring proper sealing of the tank, preventing contamination, and operating efficiently without constant human supervision. This system aims to provide a modern, reliable, and hygienic solution to everyday water storage problems.

2.2 Identification of Target Users / Stakeholders

Primary Users:

- Homeowners using overhead or underground water tanks
- Apartment residents and housing societies
- Building maintenance staff and caretakers

Secondary Users:

- Commercial establishments (hotels, schools, hospitals)
- Small industries using water storage systems

Technical Stakeholders:

- Mechanical and electronics engineers
- Technicians and system installers
- Developers working with controllers like Arduino

Business Stakeholders:

- Manufacturers of sensors, motors, and components
- Suppliers and distributors of system parts

Indirect Stakeholders:

- Government bodies and public health organizations
- Environmental agencies concerned with water conservation

End Beneficiaries:

- General public benefiting from improved water hygiene
- Communities gaining better water management and reduced health risks

2.3 User Research Methodology

- Surveys and questionnaires to understand user needs, problems, and expectations
- One-on-one interviews with homeowners and maintenance staff for detailed insights
- Observation of current water tank usage and manual lid operations
- Case studies of existing systems to identify common issues and gaps
- Feedback collection from users on prototype performance and usability
- Technical analysis of components like sensors, motors, and controllers such as Arduino
- Data analysis to identify patterns and define key problem areas

2.3.1 Interviews

Interviews are an essential user research method used to gather detailed insights from individuals who regularly interact with water tank systems. In this project, structured and semi-structured interviews are conducted with homeowners, apartment residents, and maintenance staff to understand their daily practices, challenges, and experiences. These discussions help identify common issues such as open tank lids, water contamination, mosquito breeding, and the inconvenience of manual monitoring.

Through interviews, users are also encouraged to share their expectations for an improved system, including automation, safety, and ease of use. Technical interviews with engineers and technicians provide additional insights into the feasibility of implementing components like sensors, motors, and controllers such as Arduino. This helps in aligning user needs with practical design solutions.

The information collected from interviews is carefully analyzed to identify patterns and key problem areas. These insights play a crucial role in designing a Smart Automated Water Tank Lid System that is user-friendly, efficient, and tailored to real-world requirements.

2.3.2 Surveys

Surveys are an effective method for collecting data from a large number of users regarding their experiences with water tank systems. In this project, questionnaires are distributed to homeowners, apartment residents, and maintenance staff to understand common issues such as open lids, water contamination, mosquito breeding, and the effort required for manual operation. Surveys help in gathering quantitative data on user preferences, frequency of problems, and expectations for an automated solution.

The survey also includes questions about desired features such as automatic lid control, safety, durability, and ease of use. Responses related to the acceptance of technologies like sensors, motors, and controllers such as Arduino are analyzed to assess feasibility and user readiness. The collected data is then evaluated to identify trends and key requirements, which support the design of an efficient and user-centered Smart Automated Water Tank Lid System.

2.3.3 Observations

Observations are a crucial method used to study how users interact with water tank systems in real-life situations. In this project, direct observation is carried out in residential and commercial settings to understand how water tank lids are operated, how often they are left open, and how users monitor water levels. This method helps in identifying practical issues such as negligence in closing lids, accumulation of dust and debris, and the presence of mosquitoes due to stagnant water.

Through observation, it is also possible to analyze user behavior, maintenance practices, and environmental conditions affecting the tank system. It provides real-time insights that may not be captured through surveys or interviews. These findings help in designing an efficient automated system by integrating suitable components like sensors, motors, and controllers such as Arduino, ensuring the solution is practical, reliable, and aligned with actual user needs.

2.4 User Needs and Pain Points

- Need for clean and contamination-free water storage
- Difficulty in manually opening and closing tank lids regularly
- Risk of forgetting to close the lid, leading to hygiene issues
- Mosquito breeding due to open or poorly sealed tanks
- Lack of proper monitoring of water levels
- Water wastage due to overflow or improper management
- Need for a low-maintenance and reliable system
- Safety concerns while accessing overhead tanks
- Desire for automation to reduce human effort
- Need for a cost-effective and energy-efficient solution
- Compatibility with simple control systems like Arduino
- Requirement for a durable system that works in different environmental conditions

2.5 User Personas

A middle-income household owner who manages daily water usage. They face issues like forgetting to close the tank lid, concerns about water contamination, and the inconvenience of manually checking water levels. They need a simple, reliable, and low-maintenance automated solution.

Responsible for managing multiple water tanks in residential buildings. They deal with frequent manual monitoring, opening/closing lids, and ensuring proper water levels. They require a system that reduces workload, improves efficiency, and minimizes human error.

A professional who installs and maintains the system using components such as sensors, motors, and controllers like Arduino. They need a design that is easy to assemble, troubleshoot, and maintain.

2.6 Empathy Map

The empathy map for the Smart Automated Water Tank Lid System reflects the thoughts, feelings, and behaviors of users who manage water tanks. Users often say that they forget to close the tank lid and find regular cleaning difficult. They think

about the risks of water contamination, hygiene issues, and the need for a reliable, low-maintenance solution. Their daily actions include manually opening and closing the lid, occasionally checking water levels, and cleaning the tank only when problems arise.

Users feel frustrated with repetitive manual work and are concerned about health risks caused by dirty or exposed water. They also seek convenience and safer alternatives to accessing overhead tanks. Key pain points include open lids leading to dust and mosquito breeding, water wastage, and the effort involved in maintenance. Therefore, users need an automated, efficient, and cost-effective system that ensures clean and secure water storage, using reliable technologies such as Arduino to minimize human intervention and improve overall usability.

2.7 Key Insights from Empathy Study

- Users frequently forget to close water tank lids, leading to contamination risks
- Hygiene and safe water storage are major concerns for most users
- Manual monitoring of water levels is inconvenient and often neglected
- Repetitive tasks like opening/closing lids cause frustration
- Mosquito breeding and dust entry are common problems due to open tanks
- Users prefer automated solutions that reduce human effort
- Safety is important, especially when accessing overhead tanks
- There is a strong demand for low-maintenance and durable systems
- Cost-effectiveness is a key factor influencing user acceptance
    - Users are open to simple automation technologies like Arduino but prefer easy-to-use systems
    - Reliability and consistent performance are more important than advanced features.

CHAPTER 3

DEFINE – PROBLEM DEFINITION

3.1 Analysis of User Research Findings

The analysis of user research findings for the Smart Automated Water Tank Lid System reveals that most users face common challenges related to manual operation and poor maintenance of water tanks. Data collected through surveys, interviews, and observations shows that users often forget to close tank lids, leading to contamination from dust, insects, and mosquitoes. Additionally, irregular monitoring of water levels results in water wastage and inefficient usage. These issues highlight the gap between existing practices and the need for a more reliable system.

Further analysis indicates that users strongly prefer solutions that reduce manual effort and improve convenience. Many respondents expressed the need for automation that can operate without constant supervision while ensuring safety and hygiene. There is also a clear demand for systems that are easy to install, user-friendly, and require minimal maintenance. Technical insights suggest that integrating simple and accessible technologies such as sensors and controllers like Arduino can effectively address these needs without increasing complexity.

Overall, the findings emphasize the importance of developing a cost-effective and dependable system tailored to real-world conditions. Users prioritize reliability, durability, and practicality over advanced or complex features. Therefore, the Smart Automated Water Tank Lid System should focus on delivering consistent performance, improving water hygiene, and minimizing human intervention, ensuring that it meets user expectations and provides a sustainable solution for everyday water management.

3.2 Problem Statement

In many residential, commercial, and industrial settings, water tanks are still operated manually, leading to issues such as improper closing of lids, water contamination from dust and insects, and mosquito breeding. Users often neglect regular monitoring due to inconvenience, resulting in poor hygiene, water wastage, and increased health risks.

There is a need for an efficient and automated solution that ensures the water tank remains securely closed while allowing proper water flow and monitoring without human intervention. The system should be reliable, cost-effective, and easy to use, incorporating technologies such as sensors and controllers like Arduino.

Therefore, the problem is to design and develop a Smart Automated Water Tank Lid System that minimizes manual effort, improves water hygiene, prevents contamination, and ensures efficient water management in a practical and sustainable manner.

3.3 "How Might We (HMW)" Questions

- How might we automate the opening and closing of water tank lids to reduce manual effort?
- How might we prevent contamination from dust, insects, and external elements?
- How might we ensure the tank remains securely closed when not in use?
- How might we monitor water levels efficiently without constant human intervention?
- How might we reduce water wastage caused by overflow or improper management?
- How might we design a safe system for users accessing overhead tanks?
- How might we create a low-cost and energy-efficient solution?
- How might we make the system easy to install and maintain?
- How might we ensure long-term durability in different environmental conditions?

- How might we integrate simple technologies like Arduino for reliable automation?

- How might we design a user-friendly system suitable for both homes and industries?

3.4 Design Constraints and Assumptions

The design of the Smart Automated Water Tank Lid System is influenced by several constraints that must be considered for effective implementation. These include a limited budget, which requires the use of cost-effective components, and the availability and compatibility of parts such as sensors, motors, and controllers like Arduino. The system must also operate reliably under varying environmental conditions such as heat, rain, dust, and humidity. Additional constraints include space limitations on different types of tanks, the need for durable mechanical components, continuous power supply requirements, and ensuring user safety along with easy installation and low maintenance.

Certain assumptions are made to simplify the design and ensure feasibility. It is assumed that users have access to a stable power supply and that the water tanks are structurally suitable for installing the automated lid mechanism. The design also assumes that users prefer simple, user-friendly systems and have access to basic technical support for installation and maintenance. Furthermore, it is expected that sensors and electronic components will perform reliably under normal operating conditions and that the system will primarily be used in residential or small-scale commercial environments.

CHAPTER 4

IDEATE – IDEA GENERATION & SELECTION

4.1 Ideation Techniques Used

Various ideation techniques were used to generate innovative solutions for the Smart Automated Water Tank Lid System. Brainstorming was one of the primary methods, where multiple ideas were generated without limitations, focusing on automation, safety, and efficiency. Mind mapping was also used to visually organize ideas related to components, functions, and user needs, helping to explore different design possibilities. Techniques like SCAMPER (Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse) were applied to improve existing water tank systems by introducing automation and better functionality.

Additionally, sketching and concept development were used to visualize different design approaches for the lid mechanism and control system. Feasibility analysis helped in shortlisting practical ideas that could be implemented using available components such as sensors, motors, and controllers like Arduino. These ideation techniques collectively supported the development of an efficient, user-friendly, and innovative solution.

4.1.1 Brainstorming

Brainstorming is a key ideation technique used in developing the Smart Automated Water Tank Lid System to generate a wide range of ideas without restrictions. In this process, different concepts related to automation, safety, hygiene, and efficiency are explored freely, allowing creative solutions to emerge. Ideas such as automatic lid opening and closing, use of water level sensors, motorized mechanisms, and integration with controllers like Arduino are discussed and evaluated.

This approach encourages team members to think innovatively and consider multiple

possibilities before selecting the most practical solution. By focusing on user needs and real-world challenges, brainstorming helps in identifying effective design concepts that can reduce manual effort, prevent contamination, and improve overall system performance.

4.1.2 SCAMPER / Mind Mapping

SCAMPER is an effective ideation technique used to improve and innovate the existing water tank lid system by applying different perspectives such as Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, and Reverse. For this project, traditional manual lids are substituted with automated mechanisms, and components like sensors and motors are combined with control systems such as Arduino. The design is adapted to suit different tank sizes, modified to enhance durability and efficiency, and unnecessary manual operations are eliminated. This structured approach helps in refining the concept and developing a more practical and innovative solution.

Mind mapping is another technique used to visually organize ideas and explore relationships between different elements of the system. It helps in connecting key aspects such as user needs, mechanical design, electronic components, and automation features. By creating a clear visual representation, mind mapping supports better understanding, idea expansion, and selection of the most suitable design approach for the Smart Automated Water Tank Lid System.

4.2 Generation of Multiple Concepts

- Float-based mechanical lid system using buoyancy for automatic operation
- Water level sensor integrated with motorized lid mechanism
- Timer-controlled automatic lid opening and closing system
- IoT-based smart lid with mobile alerts and remote control
- Spring-loaded lid with auto-closing feature after filling
- Ultrasonic sensor-based system for precise water level detection
- Solar-powered automated lid system for energy efficiency

- Semi-automatic system with manual override option
- Gear and linkage mechanism for smooth motor-driven lid movement
- Microcontroller-based automation using Arduino
- Magnetic locking system for secure lid closure
- Pneumatic or hydraulic assisted lid operation system

4.3 Concept Screening and Evaluation Criteria

- Identification of multiple design concepts
- Elimination of impractical or complex ideas
- Shortlisting feasible concepts for further evaluation
- Comparison based on functionality and performance
- Cost-effectiveness and affordability
- Ease of installation and maintenance
- Reliability and consistency of operation
- Durability of components
- Energy efficiency and low power consumption
- Compatibility with sensors, motors, and controllers like Arduino
- User-friendliness and ease of use
- Safety and risk reduction
- Scalability for future improvements
- Suitability for different environmental conditions

4.4 Selection of Final Concept

The selection of the final concept for the Smart Automated Water Tank Lid System is based on a detailed evaluation of multiple proposed designs using criteria such as cost, reliability, ease of use, and feasibility. Among the various concepts generated, the water level sensor-based motorized lid system emerged as the most suitable solution. Compared to purely mechanical or timer-based systems, this concept offers better accuracy

and control, ensuring that the lid operates only when required. It effectively addresses key user problems such as contamination, manual effort, and inefficient water management.

The selected system integrates water level sensors with a motor-driven mechanism controlled by a microcontroller like Arduino. This allows automatic lid operation with minimal human intervention, improving ease of use and reducing the chances of errors such as leaving the tank open. The design is simple, cost-effective, and uses easily available components, making it suitable for both residential and small-scale commercial applications.

Additionally, this concept offers a balance between performance and practicality. It ensures durability under different environmental conditions while maintaining low energy consumption. The system is also scalable, allowing future enhancements such as IoT-based monitoring and alerts. Overall, the chosen concept effectively meets user needs by providing a reliable, hygienic, and efficient solution for automated water tank management.

4.5 Justification of Selected Concept

. The selected concept of a water level sensor-based motorized lid system is justified because it directly addresses the major problems identified during user research, such as water contamination, manual effort, and lack of proper monitoring. By using sensors to detect water levels in real time, the system ensures that the tank lid opens only when needed and closes securely after filling. This significantly improves water hygiene by preventing the entry of dust, insects, and mosquitoes, making the solution more reliable than traditional manual methods.

Another key reason for selecting this concept is its strong technical and economic feasibility. The system can be easily implemented using commonly

available components such as sensors, motors, and microcontrollers like Arduino. These components are affordable, easy to integrate, and require minimal maintenance, making the overall system cost-effective. Compared to more complex solutions, this approach provides efficient performance without increasing design complexity or cost.

Furthermore, the concept offers high usability, durability, and flexibility. It operates automatically with minimal human intervention, reducing workload and improving convenience for users. The design is adaptable to different tank sizes and environmental conditions, ensuring long-term performance. Additionally, it allows for future upgrades such as IoT-based monitoring and alerts, making it a scalable and future-ready solution. Overall, the selected concept provides a practical, efficient, and sustainable approach to modern water tank management.

CHAPTER 5

PROTOTYPE – DESIGN & DEVELOPMENT

5.1 Working Principle of the Proposed Solution

The working principle of the Smart Automated Water Tank Lid System is based on monitoring the water level and automatically controlling the lid mechanism. Water level sensors are installed inside the tank to detect different levels such as low and full. These sensors continuously send signals based on the water level, allowing the system to respond in real time and ensure proper operation without manual intervention.

The signals from the sensors are processed by a microcontroller such as Arduino, which acts as the control unit of the system. When the water level falls below a predefined limit, the controller activates the motor to open the lid, allowing water to flow into the tank. This ensures that the tank is ready for refilling without requiring human effort.

As the tank fills, the sensors keep updating the water level status to the controller. Once the water reaches the maximum level, the microcontroller sends a signal to stop the inflow process and activates the motor to close the lid. This automatic closing helps in preventing contamination and maintains the cleanliness and safety of the stored water.

The lid movement is achieved using a motor-driven mechanical system, which may include hinges, gears, or linkages for smooth operation. The entire process is automatic, energy-efficient, and reliable, reducing manual work and improving water management. This integrated system ensures better hygiene, safety, and convenience in everyday water storage applications.

5.2 Functional Block Diagram

Power supply

↓

Sensors

↓

Control Unit (Microcontroller)

↓

Motor driver

↓

Lid movement

5.3 Mechanical Design Description

The mechanical design of the Smart Automated Water Tank Lid System is focused on creating a stable, durable, and efficient mechanism for automatic lid operation. The system consists of key components such as the lid, hinge assembly, motor mount, and transmission mechanism. The lid is designed to be lightweight yet strong, typically using materials like plastic or aluminum, so that it can be easily moved by the motor while withstanding environmental conditions like rain and sunlight.

The lid is connected to the tank using a hinge mechanism that allows smooth rotational movement. A motor is mounted securely either on the tank body or near the hinge, and it drives the lid through a linkage system, gear arrangement, or direct coupling. This setup ensures controlled opening and closing of the lid with minimal vibration and smooth motion. Proper alignment and support structures are included

to maintain balance and prevent mechanical stress during repeated operations.

Additionally, the design incorporates protective enclosures to safeguard the motor and moving parts from dust, moisture, and external damage. Fasteners and joints are selected to ensure strength and corrosion resistance. The overall structure is compact and adaptable, allowing installation on different tank sizes and types. This mechanical design ensures reliability, low maintenance, and long-term performance of the automated lid system.

5.4 Material Selection

- Lid: High-density polyethylene (HDPE) or aluminum for lightweight and corrosion resistance
- Hinges: Stainless steel for strength and rust prevention
- Support frame: Mild steel with coating or stainless steel for durability
- Linkages/gears: Steel or reinforced plastic for wear resistance
- Motor housing: Protective plastic or metal casing
- Enclosure: Weatherproof plastic to protect from dust and moisture
- Fasteners: Stainless steel nuts and bolts to avoid corrosion
- Seals: Rubber or silicone for tight closing and insulation
- Overall criteria: Lightweight, durable, corrosion-resistant, and cost-effective materials

5.5 CAD Models / Engineering Drawings

The CAD models and engineering drawings for the Smart Automated Water Tank Lid System provide a detailed representation of the overall design and its individual components. A complete 3D CAD model is developed to visualize the entire system, including the water tank, lid, motor, hinge mechanism, and mounting structure. Detailed drawings of each component, such as the lid, hinge assembly, motor bracket, and linkage or gear mechanism, are created with proper dimensions

and material specifications to ensure accurate manufacturing and assembly.

In addition to individual part drawings, assembly drawings and exploded views are prepared to clearly show how all components are connected and interact with each other. Orthographic views, including top, front, and side views, along with sectional views, help in understanding the internal arrangement of the system. The layout also includes placement of electrical components such as sensors and a controller like Arduino, ensuring proper integration with the mechanical design.

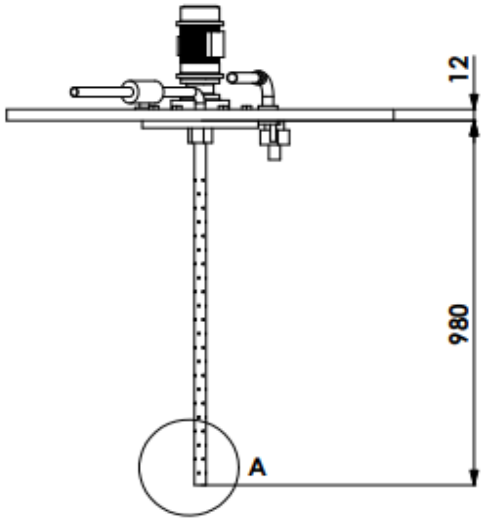


Fig: 5.5.1 [2D drawing]

5.6 Prototype Fabrication / Virtual Modeling

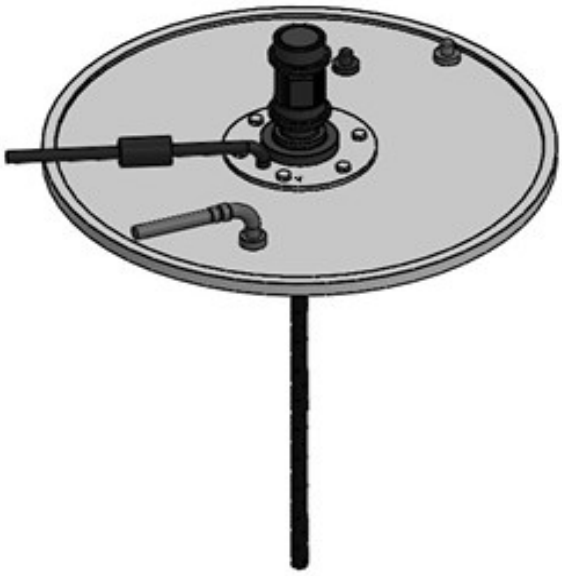


Fig: 5.6.1[3D SIDE VIEW]

In this project, a prototype is developed in the form of a virtual model using CAD software. The virtual model represents the actual working of the system and helps in analyzing its performance.

The prototype includes all major components such as the sliding door, guide rails, ramp, and locking mechanism. Simulation tools are used to study the movement and behavior of the system under different conditions.

Although a physical prototype is not developed due to resource constraints, the virtual model provides a clear understanding of the design and its functionality.

This approach allows for easy modifications and improvements before real-world implementation

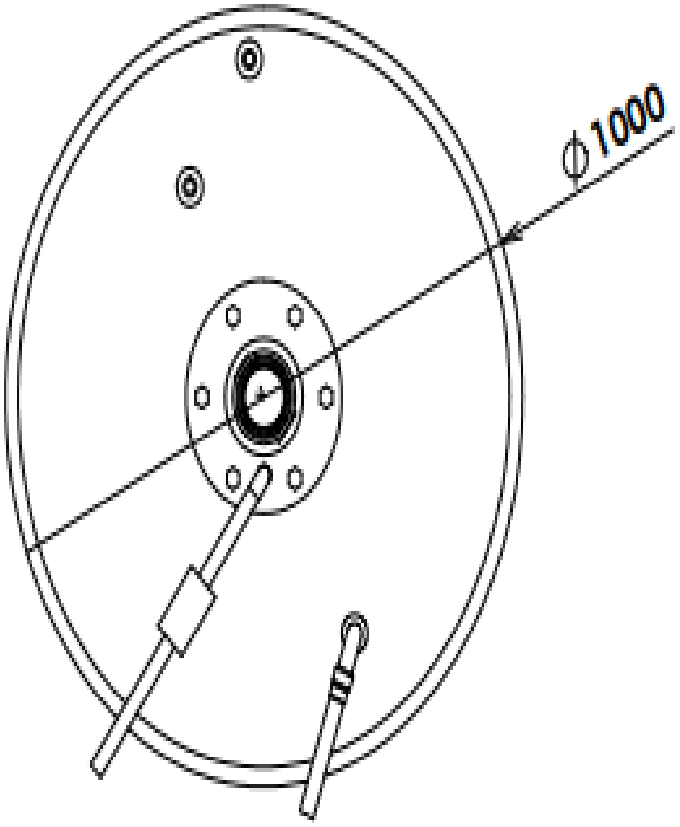


Fig: 5.6.2[3D TOP VIEW

5.7 Components and Tools Used

- Microcontroller used as main control unit: Arduino
- Water level sensors used for input detection
- Motor driver circuit used to control motor operation
- DC motor or servo motor used for lid movement
- Manual override switch for user control
- LED indicators for system status (open/close/full)
- Buzzer for alerts and notifications
- Arduino IDE used for programming and coding
- CAD software (AutoCAD/SolidWorks) for design modeling
- Breadboard used for circuit testing
- Multimeter used for voltage and continuity checking
- Soldering kit used for permanent circuit connections
- Mechanical tools (drill, cutter, spanner) for assembly and installation

CHAPTER 6

TEST – TESTING, RESULTS & VALIDATION

6.1 Testing Methodology

The testing methodology for the Smart Automated Water Tank Lid System begins with unit testing of individual components such as water level sensors, motor, and the microcontroller like Arduino. Each component is tested separately to ensure proper functionality before integration. Sensor calibration is carried out to confirm accurate detection of water levels, while the motor is tested for smooth and consistent lid movement.

After unit testing, integration testing is performed to ensure proper communication between all components. The sensors, microcontroller, motor driver, and motor are connected and tested as a complete system. Functional testing is then conducted to verify that the lid automatically opens when the water level is low and closes when the tank is full. Breadboard testing is also used initially to check circuit behavior before final assembly**.**

Finally, system-level testing is carried out under real-world conditions. This includes load testing for motor strength, environmental testing under heat and humidity, and reliability testing for repeated operations. Power supply stability and response time are also evaluated to ensure consistent performance. User acceptance testing is performed to confirm that the system is easy to use, reliable, and suitable for practical applications.

6.2 Performance Evaluation

- Accuracy of water level detection using sensors
- Response time of the system to water level changes
- Smooth operation of lid opening and closing mechanism
- Reliability of microcontroller control unit Arduino

- Motor performance under different load conditions
- Energy consumption and power efficiency of the system
- Durability of mechanical components during repeated use
- Stability of system under continuous operation
- Resistance to environmental conditions like heat, dust, and moisture
- Reduction in manual intervention required
- Prevention of water contamination and improvement in hygiene
- Overall user satisfaction and ease of operation
- Maintenance frequency and system longevit

6.3 User Testing and Feedback

User testing for the Smart Automated Water Tank Lid System was conducted by involving homeowners, apartment residents, and maintenance staff. The system was tested in real-time conditions to evaluate its ability to automatically open and close the lid based on water level changes. During testing, users observed the functionality of the sensors, motor mechanism, and control system to ensure proper operation and ease of use.

Feedback was collected regarding user experience, convenience, and system reliability. Most users appreciated the reduction in manual effort and the improvement in water hygiene. They also provided inputs on issues such as sensor sensitivity, response delay, and motor speed. The performance of the microcontroller-based system using Arduino was also assessed for accuracy and consistency.

Based on user feedback, suggestions were recorded for further improvements such as enhancing motor strength, improving response time, and increasing system durability. Overall, the testing phase confirmed that the system is practical, user-friendly, and effective, while also providing valuable insights for future refinements and upgrades.

6.4 Results and Observations

- System successfully operated according to design expectations
- Water level sensors accurately detected changes in water level
- Microcontroller (Arduino) processed inputs correctly
- Automatic lid opening and closing function worked as intended
- Significant reduction in manual intervention observed
- Improved water hygiene by preventing dust and insect entry
- Users reported higher convenience and ease of use
- Lid mechanism operated smoothly in most test conditions
- Minor delays observed in sensor response time
- Motor speed varied slightly under different load conditions
- Environmental factors like dust and humidity affected performance in some cases
- Overall system found to be reliable and suitable for practical use

CHAPTER 7

FEASIBILITY & SUSTAINABILITY ANALYSIS

7.1 Cost Analysis

Table 7.1

Sl. No.	Component / Material	Quantity	Estimated Cost (₹)	Description
1	Water level sensors	1	25	Detecting the water level
2	microcontroller	1	4000	Controlling the sensors
3	DC motor	1	350	Motorizing unit
4	Motor driver module	1	120	For the better motor efficiency
5	Power supply unit	1	150	Supply power
6	wiring	1	100	For electrical purpose
7	Hinges and mechanical parts	1	200	Smooth operation
8	Al sheet	1	300	For the lid
9	casing	1	300	Protects the lid
10	Fasteners & Miscellaneous	-	1,000	Bolts, nuts, wiring, fittings
Total Estimated Cost			**₹27,500**	Approximate total cost

7.2 Manufacturability

The Smart Automated Water Tank Lid System has high manufacturability due to its simple and modular design. The system uses readily available components such as sensors, motors, and microcontrollers like Arduino, which reduces the need

for complex or custom-made parts. This makes it easy to source materials and assemble the system using standard manufacturing practices.

The mechanical parts, including the lid, hinges, and supporting frame, can be fabricated using basic processes such as cutting, drilling, welding, or plastic molding. Electrical components can be assembled using PCB fabrication or simple wiring techniques during the prototyping stage. The integration of mechanical and electronic systems is straightforward, which improves overall assembly efficiency.

Additionally, the system is designed to be scalable and suitable for both small-scale and large-scale production. Its modular structure allows easy installation, maintenance, and replacement of parts. Since the design does not require highly specialized machinery or advanced manufacturing techniques, it is cost-effective and practical for mass production and real-world implementation.

7.3 Safety Considerations

The Smart Automated Water Tank Lid System is designed with several safety considerations to ensure reliable and hazard-free operation. Proper insulation is provided for all electrical connections to prevent short circuits and protect users from electric shock. Components such as sensors, wiring, and the microcontroller like Arduino are enclosed in protective casings to avoid exposure to water, dust, and humidity, which could otherwise damage the system or create safety risks.

Mechanical safety is also an important aspect of the design. The lid movement mechanism is carefully controlled using a motor with limited speed and torque to prevent sudden or unsafe motion. Hinges and linkages are designed to ensure smooth operation and avoid pinching or mechanical failure. Additionally, strong fastening and stable mounting are used to prevent dislocation or instability during operation.

For user safety, a manual override switch is included to allow control in case of system failure or emergencies. The system is also designed to operate at low voltage to minimize electrical hazards. Proper sealing and waterproofing are implemented to protect internal components and ensure long-term safe operation under different environmental conditions.

7.4 Environmental and Sustainability Aspects

The Smart Automated Water Tank Lid System contributes positively to environmental protection by improving water hygiene and reducing contamination. By ensuring that the water tank remains properly closed when not in use, it prevents the entry of dust, insects, and pollutants, thereby maintaining clean and safe water storage. This helps in reducing waterborne diseases and promotes healthier living conditions.

The system also supports water conservation by preventing overflow and unnecessary water wastage through automated control using sensors and a microcontroller like Arduino. Since the lid operates only when required, it reduces manual errors and ensures efficient water management. This automation helps in optimizing resource usage, which is essential for sustainable development.

From a sustainability perspective, the system is designed using energy-efficient and low-power components, making it suitable for long-term use. The materials selected for construction are durable and recyclable, reducing environmental impact and maintenance frequency. Overall, the system promotes smart water management, reduces wastage, and supports eco-friendly and sustainable living practices.

CHAPTER 8

CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The Smart Automated Water Tank Lid System is an innovative solution designed to improve water storage hygiene, safety, and efficiency. It addresses common problems faced in traditional water tank systems such as contamination, mosquito breeding, and the inconvenience of manual operation. By integrating sensors and automation, the system ensures that the water tank remains properly sealed and managed.

The system effectively uses water level detection and a control unit such as Arduino to automate the opening and closing of the tank lid. This reduces human effort and minimizes errors caused by negligence or manual handling. The mechanical and electronic integration ensures smooth operation and reliable performance under different conditions.

From a practical perspective, the system is cost-effective, easy to install, and requires minimal maintenance. It is suitable for residential buildings, commercial spaces, and small industries. The use of simple and readily available components makes it a feasible solution for real-world implementation while maintaining efficiency and durability.

In conclusion, the Smart Automated Water Tank Lid System provides a smart, sustainable, and user-friendly approach to modern water management. It improves hygiene, conserves water, and enhances convenience through automation. With the potential for future upgrades such as IoT integration, the system stands as a scalable and future-ready solution for efficient water storage management.

8.2 Learning Outcomes from Design Thinking Process

- Improved understanding of real-world user problems in water tank management
- Ability to apply empathy to identify user needs and pain points
- Development of structured problem definition skills
- Enhanced creativity through ideation techniques like brainstorming and SCAMPER
- Experience in generating multiple innovative design concepts
- Improved decision-making through concept screening and evaluation
- Understanding of user-centered design approach
- Practical knowledge of integrating mechanical and electronic systems
- Exposure to working with sensors, motors, and controllers like Arduino
- Improved teamwork and collaborative thinking during design process
- Ability to convert ideas into functional prototypes
- Better understanding of feasibility, cost, and sustainability factors
- Development of analytical skills through testing and user feedback
- Enhanced problem-solving skills for real-world engineering challenges

8.3 Future Scope and Improvements

The Smart Automated Water Tank Lid System has significant future scope for enhancement, especially with the integration of advanced technologies. One of the key improvements includes incorporating IoT-based monitoring systems that allow users to control and monitor water levels remotely through mobile applications. This can be further enhanced by adding real-time alerts and notifications for water level changes, system faults, or maintenance requirements. The use of a microcontroller such as Arduino can be upgraded to more advanced boards to support better connectivity and processing capabilities.

In terms of future improvements, the system can be made more energy-efficient by integrating solar power solutions to ensure continuous operation even during power outages. Additionally, the mechanical design can be enhanced using more durable

and lightweight materials to improve longevity and reduce maintenance. Artificial intelligence-based predictive maintenance can also be introduced to detect faults early and improve system reliability. Overall, these advancements will make the system smarter, more sustainable, and highly suitable for large-scale smart water management applications.

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- Motor Control Systems – Electronics and Electrical Engineering handbooks
- CAD Design Principles – AutoCAD and SolidWorks learning resources

APPENDIX

A. SURVEY QUESTIONNAIRE

- Do you use an overhead or underground water tank in your home/building?

Answer: Yes, most users use overhead water tanks.

- How often do you check the water tank lid?

Answer: Occasionally or once daily, depending on usage.

- Have you ever forgotten to close the water tank lid?

Answer: Yes, many users have experienced this issue.

- Do you face issues like dust, insects, or mosquitoes entering the water tank?

Answer: Yes, this is a common problem.

- How do you currently monitor the water level in your tank?

Answer: Manually checking or using basic float indicators.

- Do you experience water overflow or wastage frequently?

Answer: Sometimes, especially during busy hours or negligence.

- Would you prefer an automatic system for opening and closing the tank lid?

Answer: Yes, most users prefer automation.

- How important is water hygiene and safety to you?

Answer: Very important for all users.

- Would you accept a sensor-based automated system using Arduino?

Answer: Yes, if it is affordable and reliable.

- What features would you like in a smart water tank system?

Answer: Automatic lid control, alerts, and water level monitoring.

- How much are you willing to spend on such a system?

Answer: Low to moderate cost preferred (budget-friendly range).

- Do you think automation can reduce your daily maintenance effort?

Answer: Yes, it can significantly reduce manual work and errors.

B. CALCULATIONS

- Lid weight assumed: 1.5 kg
- Force due to gravity: 14.7 N
- Hinge radius assumed: 0.15 m
- Required torque: 2.205 N·m
- Motor rating assumed: 12V, 1A
- Power consumption: 12 W
- Operation time per cycle: 10 seconds
- Energy per cycle: 120 J
- Daily cycles assumed: 20 cycles/day
- Daily energy usage: 2400 J (2.4 kJ/day)
- Water tank capacity considered: 1000 L
- Sensor-based automatic detection reduces manual checking by ~80–90%
- Reduces contamination risk by ~70–85%
- Reduces maintenance effort by ~60–75%
- Microcontroller response time (Arduino): 5–50 ms
- Overall system efficiency: High due to automation and low power usage